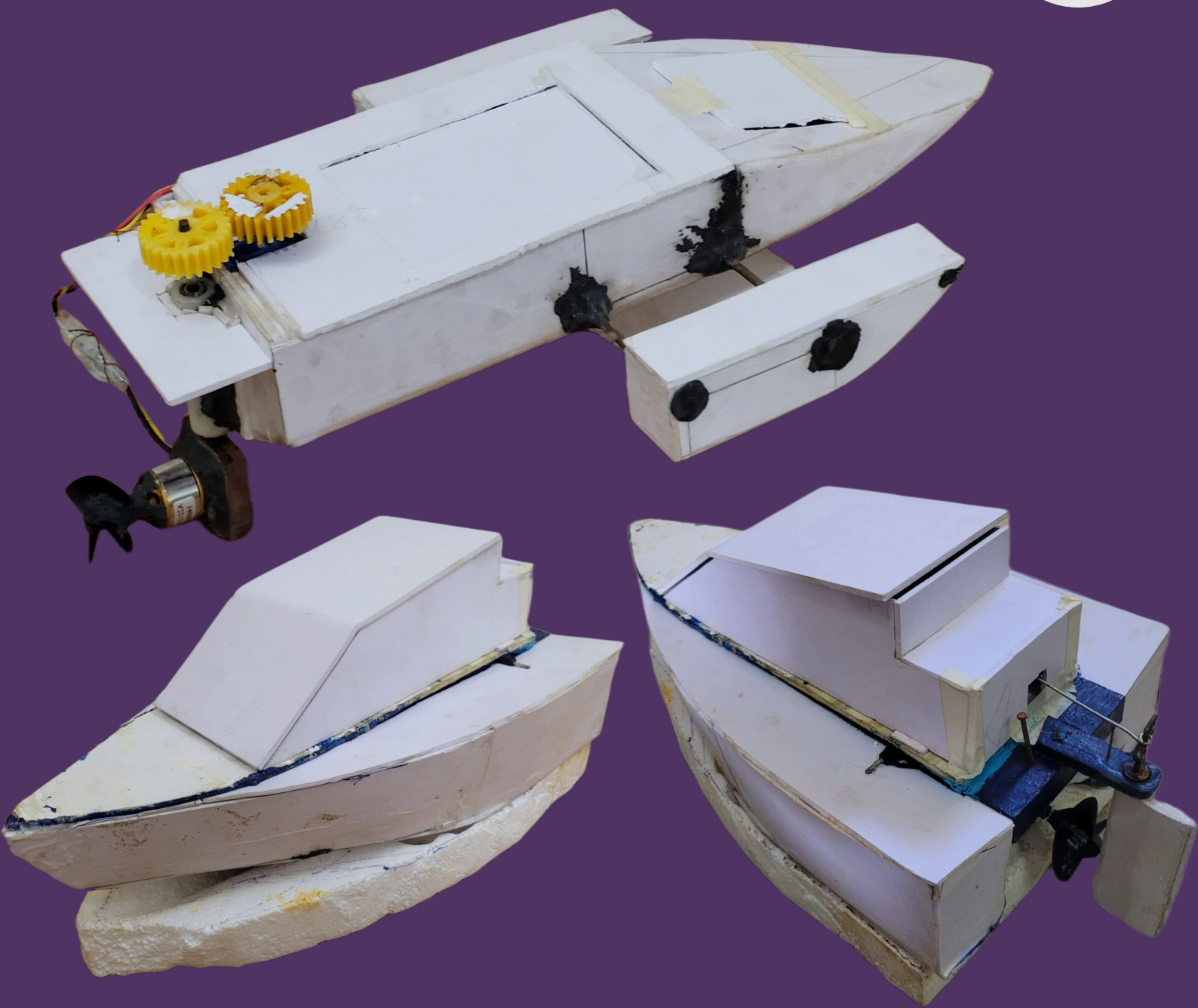




REMOTELY OPERATED BOATS

IITM & SM24



IN SEARCH OF INNOVATION

BuildNEO: A student-led research and development initiative at the Indian Maritime University, Visakhapatnam, focused on innovation and practical skill development in naval architecture. Founded by ambitious alumni, it thrives on a network of past and present students collaborating on working prototypes and research. The initiative features a dynamic leadership team, including roles in external communications, electronics, technical documentation, coding, CAD modeling, fabrication, additive manufacturing and social media. By promoting project flexibility and creativity, BuildNEO prepares members for academic and industrial challenges, aiming to become a hub of talent and innovation in the maritime industry.

TANK PEP

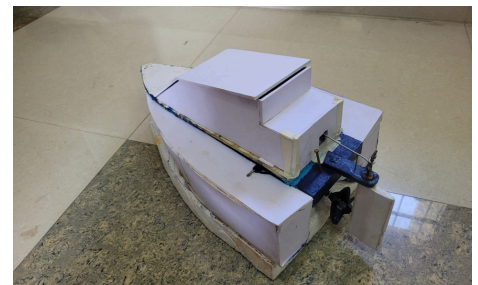
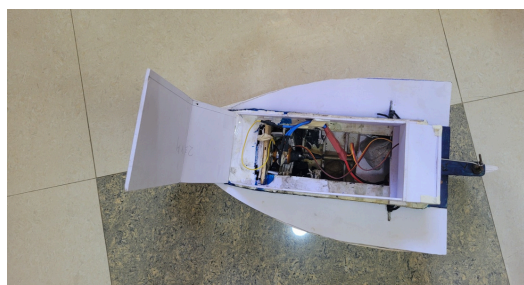
The project aims to develop a mono-hull boat system equipped with radar technology for navigation and obstacle detection. It features a single propulsion unit powered by a BLDC motor, which is linked to the radar-propeller system via a shaft mechanism. This integration ensures efficient radar-assisted navigation while optimizing propulsion performance. The system is designed for reliable operation and accurate radar detection, enhancing control and safety during use.

OBJECTIVE

1. Integration of Radar Technology – Enable navigation and obstacle detection for improved situational awareness.
2. Efficient Propulsion System – Utilize a BLDC motor to drive the boat with optimal performance and energy efficiency.
3. Shaft-Linked Radar-Propeller Mechanism – Ensure seamless coordination between propulsion and radar functionality.
4. Reliable Operation – Develop a system that maintains stability and efficiency under various conditions.
5. Accurate Obstacle Detection – Enhance safety by enabling precise radar-based detection of obstacles.
6. Improved Control and Maneuverability – Ensure smooth and responsive navigation for effective operation.

SPECIFICATIONS

1. Length: 48 cm
2. Breadth: 22 cm
3. Draft: 5 cm
4. Max Speed: 1.8 knots
5. Range: 100 m at 1 knot
6. Endurance: 1 hour
7. Capacity: BLDC motor, LiPo battery, receiver, ESC





This project aims to design and develop a high-performance model speed boat that uses an azimuthal propulsion mechanism to enhance maneuverability and efficiency. The propulsion system, driven by a BSLV motor and a gear-based transmission, enables precise thrust vectoring with a 90-degree rotation and ± 45 -degree adjustments for optimal directional control. Stability is ensured through a twin dummy catamaran hull structure, while a servo-driven steering system facilitates accurate navigation. The integration of these advanced components seeks to improve hydrodynamic performance, propulsion efficiency, and overall agility in waterborne applications.

METHODOLOGY

Design and simulation using CAD software
 Prototype development with azimuthal propulsion and steering systems
 Experimental testing in a controlled water environment
 Data analysis and optimization using CFD and statistical methods

FEATURES

1. Trispeed System with 1400KV Brushless Motor
 - Three speed levels powered by a high-RPM (1400KV) brushless motor for efficient propulsion.
2. Azimuthal Propulsion with Servo Control
 - Motor can rotate $\pm 45^\circ$ and up to 90° using a servo, allowing precise directional control without a rudder.
3. Dual Dummy Catamaran Hulls for Stability
 - Two side hulls provide balance and reduce tilting, ideal for smooth navigation in water.
4. Gear Mechanism for Power Transmission
 - Gears optimize motor output and control rotation for propulsion and direction.
5. Highly Maneuverable and Stable Design
 - Combines speed control, precise steering, and stable hull structure for efficient water movement.

SPECIFICATIONS

1. Length: 48 cm
2. Breadth: 28.5 cm
3. Draft: 4.5 cm
4. Max Speed: 3.5 knots
5. Range: 180 m at 1.8 knots
6. Endurance: 25 minutes
7. Capacity: BLDC motor, LiPo battery, receiver, ESC, dual gear

